

Md Ashiqur Rahman

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Research Interest

Designing robust and physically grounded AI systems, spanning geometric and scale-equivariant deep learning, neural operators for scientific computing (AI for Science), and multimodal models (VLMs/VLAs) for embodied decision-making and real-world perception.

Education

Purdue University **West Lafayette, IN**
Doctor of Philosophy in Computer Science 2021–

Advisor: Raymond A. Yeh

[Awarded Bilsland Dissertation Fellowship \(2026–2027\)](#)

Highly competitive, nomination-based fellowship for top final-year PhD candidates (1–2 per department).

Purdue University **West Lafayette, IN**
Master of Science in Computer Science 2021–2025

Bangladesh University of Engg & Tech **Dhaka, Bangladesh**
Bachelor of Science in Computer Science and Engineering 2015–2019

Advisor: Md. Shamsuzzoha Bayzid

Professional Employment

Purdue University **West Lafayette, IN**
Graduate Assistant in the Department of Computer Science 2021–present

Eli Lilly **Indianapolis, IN**
Research Intern Summer 2026

Advanced Intelligence Team

- Developed an edge-deployed agentic-vision system for monitoring pharmaceutical manufacturing.

Autodesk **San Francisco, CA**
Research Intern Summer 2025

Manager: Robert Giaquinto

- Developed robust VLMs for 3D CAD model generation for engineering.

NVIDIA **Santa Clara, CA**
Research Intern Summer & Fall 2023

Manager: Anima Anandkumar

- Developed techniques for multi-physics pretraining for foundation models and large-scale weather simulation.

United International University **Dhaka, Bangladesh**
Lecturer in the Department of Computer Science 2019–2021

Publications

Conference

- [C1] **Md Ashiqur Rahman**, Jun Hao Lim, Jeremiah Jiang, Teck-Yian Lim, and Raymond A. Yeh. Tunable Soft Equivariance with Guarantees. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE/CVF, 2026. [\[arXiv\]](#) [\[Code\]](#).
- [C2] **Md Ashiqur Rahman**, Chiao-An Yang, Michael N. Cheng, Jun Hao Lim, Jeremiah Jiang, Teck-Yian Lim, and Raymond A. Yeh. Local Scale Equivariance with Latent Deep Equilibrium Canonicalizer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. [\[arXiv\]](#) [\[Code\]](#).
- [C3] Tinghan Yang, **Md Ashiqur Rahman**, and Raymond A. Yeh. CLIPSym: Delving into Symmetry Detection with CLIP. In *International Conference on Computer Vision (ICCV)*, 2025. [\[arXiv\]](#) [\[Code\]](#).
- [C4] **Md Ashiqur Rahman** and Raymond A. Yeh. Group Downsampling with Equivariant Anti-aliasing. In *International Conference on Learning Representations (ICLR)*, 2025. [\[OpenReview\]](#) [\[Code\]](#).

- [C5] **Md Ashiqur Rahman**, R. J. George, M. Elleithy, D. Leibovici, Z. Li, B. Bonev, C. White, J. Berner, R. A. Yeh, J. Kossaifi, K. Azizzadenesheli, and A. Anandkumar. Pretraining Codomain Attention Neural Operators for Solving Multiphysics PDEs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. [[arXiv](#)] [[Code](#)].
- [C6] **Md Ashiqur Rahman** and Raymond A. Yeh. Truly Scale-Equivariant Deep Nets with Fourier Layers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. [[arXiv](#)] [[Code](#)].
- [C7] **Md Ashiqur Rahman**, Abdullah Aman Tutul, and A. B. M. Alim Al Islam. Solving The Maze of Diagnosing Parkinson’s Disease based on Portable EEG Sensing to be Adaptable to Go In-The-Wild. *Proceedings of the 7th International Conference on Networking, Systems and Security*, 2020. (Best Paper Award) [[DOI](#)].
- [C8] Akm Ashiquzzaman, Abdul Kawsar Tushar, **Md Ashiqur Rahman**, and Farzana Mohsin. An Efficient Recognition Method for Handwritten Arabic Numerals Using CNN with Data Augmentation and Dropout. *Data Management, Analytics and Innovation*, 2018. [[DOI](#)].

Journal

- [J1] Hrishikesh Viswanath, **Md Ashiqur Rahman**, Abhijeet Vyas, Andrey Shor, Beatriz Medeiros, Stephanie Hernandez, Suhas Eswarappa Prameela, and Aniket Bera. Neural Operator: Is data all you need to model the world? An insight into the paradigm of data-driven scientific ML. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pages 1–18, 2026. [[IEEE TPAMI](#)] [[arXiv](#)].
- [J2] **Md Ashiqur Rahman**, Abdullah Aman Tutul, Mahfuza Sharmin, and Md. Shamsuzzoha Bayzid. BEENE: deep learning-based nonlinear embedding improves batch effect estimation. *Bioinformatics*, 2023. [[DOI](#)] [[Code](#)].
- [J3] **Md Ashiqur Rahman**, Manuel A. Florez, Anima Anandkumar, Zachary E. Ross, and Kamyar Azizzadenesheli. Generative Adversarial Neural Operators. *Transactions on Machine Learning Research*, 2022. [[TMLR](#)] [[Code](#)].
- [J4] **Md Ashiqur Rahman**, Zachary E. Ross, and Kamyar Azizzadenesheli. U-NO: U-shaped Neural Operators. *Transactions on Machine Learning Research*, 2022. [[TMLR](#)] [[Code](#)].
- [J5] **Md Ashiqur Rahman**, Abdullah Aman Tutul, Sifat Muhammad Abdullah, and Md. Shamsuzzoha Bayzid. CHAPAO: Likelihood and hierarchical reference-based representation of biomolecular sequences and applications to compressing multiple sequence alignments. *PLoS ONE*, 2022. [[PLoS ONE](#)] [[Code](#)].

Under Review

- [S1] Kaivalya Agrawal, **Md Ashiqur Rahman**, Raymond A. Yeh, and Zachary Kingston. SkipVLA: Skipping VLA Steps with Classical Planning for Fast Robot Manipulation. Under Review. [[arXiv](#)].
- [S2] Hrishikesh Viswanath, **Md Ashiqur Rahman**, Chi Lin, Damon Conover, and Aniket Bera. HessianForge: Scalable LiDAR reconstruction with Physics-Informed Neural Representation and Smoothness Energy Constraints. Under Review. [[arXiv](#)] [[Code](#)].

Patents

- [P1] Aniket Bera, Rashmi Bhaskara, **Md Ashiqur Rahman**, and Hrishikesh Viswanath. Physics-informed adaptive fourier neural interpolation operator for synthetic frame generation, January 9 2025. US Patent App. 18/758,927 [[Patent](#)].

Awards & Recognition

- Bilsland Dissertation Fellowship** 2026–2027
 - Competitive nomination-based fellowship for top final-year PhD candidates (1–2 per department).
- ACCESS Discover Project Award** 2026–2027
 - Awarded 150,000 computing credits (~\$10K–15K equivalent) in national cyberinfrastructure resources to support scientific discovery.

Teaching Experience

Purdue University

West Lafayette, IN

Teaching Assistant - CS587: Foundations of Deep Learning

Spring 2024

- Co-designed and graded course assignments and exam questions involving equivariant models, optimizing deep neural networks, generative models, and optimization layers.

Purdue University

West Lafayette, IN

Teaching Assistant - CS373: Data Mining and Machine Learning

Fall 2022, Spring 2023

- Co-designed course assignments and exam questions involving fundamentals of machine learning, perceptron, SVM, learning theory, and data privacy.

Purdue University

West Lafayette, IN

Teaching Assistant - CS334: Fundamentals of Computer Graphics

Fall 2021, Fall 2023

- Co-designed course assignments and exam questions involving GPU programming, ray tracing, texture mapping, and procedural modeling.

United International University

Dhaka, Bangladesh.

Lecturer

2019 - 2021

- Redesigned and instructed courses, including Data Structures and Algorithms, Simulation and Modeling, Intro to Computer Graphics, and Computer Networking.

Services

Professional Service

Conference Reviewer:

- Neural Information Processing Systems
- International Conference on Machine Learning
- International Conference on Learning Representations
- Computer Vision and Pattern Recognition

Journal Reviewer:

- IET Computer Vision
- IEEE Transactions on Neural Networks and Learning Systems

Research with Undergraduates

Michael Cheng

Undergraduate, Purdue, Fall 2024 – Fall 2025. Conducting undergraduate research under the supervision of **Professor Raymond A. Yeh**. As a mentor, I receive weekly updates, provide feedback, set project directions, and offer hands-on guidance in project development and implementation.

Research topic: *Equivariant machine learning for computer vision.*

Changxiang Gao

Undergraduate, Purdue, Summer 2024 – Fall 2024. Conducted undergraduate research under the supervision of **Professor Raymond A. Yeh**. As a mentor, I received weekly updates, defined project goals, and provided hands-on guidance for implementation.

Research topic: *Any-resolution image classification.*

Anugu Arun Reddy

Visiting Scholar, Purdue, Summer 2024 under the supervision of **Professor Aniket Bera**. As a mentor, I defined the project scope, received weekly updates, and guided the implementation process.

Research topic: *Human motion generation.*